In [ ]:
```
# About Walmart
# Walmart is an American multinational retail corporation that operates a chain of supercenters, discount departmental stores,
# and grocery stores from the United States. Walmart has more than 100 million customers worldwide.

# Business Problem
# The Management team at Walmart Inc. wants to analyze the customer purchase behavior (specifically, purchase amount) against the customer's gender
# and the various other factors to help the business make better decisions. They want to understand if the spending habits differ
# between male and female customers: Do women spend more on Black Friday than men? (Assume 50 million customers are male and 50 million are female).
```

In [41]:
```
## Importing Libraries
## We import essential Python libraries required for data analysis, visualization, and manipulation.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
from scipy.stats import norm

pd.set_option('display.max_columns', None)
pd.set_option('display.max_colwidth', None)
pd.set_option('display.width', 5000)
```

In [42]:
```
## 1. Import the dataset and do usual data analysis steps like checking the structure & characteristics of the dataset.
## The Aerofit-Treadmill dataset is loaded into a pandas DataFrame for analysis.

df_walmart = pd.read_csv(r'C:\Users\amit.sehrawat\Python_Practise\Datasets\walmart_data.csv')
```

In [43]:
```
## 1.a. View the data
# First 5 rows
df_walmart.head(5)
```

Out[43]:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	1000001	P00069042	F	0-17	10	A	2	0	3	8370
1	1000001	P00248942	F	0-17	10	A	2	0	1	15200
2	1000001	P00087842	F	0-17	10	A	2	0	12	1422
3	1000001	P00085442	F	0-17	10	A	2	0	12	1057
4	1000002	P00285442	M	55+	16	C	4+	0	8	7969

In [44]:
```
# Last 5 rows
df_walmart.tail(5)
```

Out[44]:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
550063	1006033	P00372445	M	51-55	13	B	1	1	20	368
550064	1006035	P00375436	F	26-35	1	C	3	0	20	371
550065	1006036	P00375436	F	26-35	15	B	4+	1	20	137
550066	1006038	P00375436	F	55+	1	C	2	0	20	365
550067	1006039	P00371644	F	46-50	0	B	4+	1	20	490

In [45]:
```
# Random rows
df_walmart.sample(5)
```

Out[45]:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
134533	1002786	P00288942	M	26-35	4	A	2	0	1	15844
219133	1003792	P00339842	M	26-35	6	A	1	0	1	19083
86841	1001390	P00119642	M	36-45	16	C	3	1	1	19044
277356	1000777	P00251542	M	18-25	19	B	1	0	11	4492
533045	1004054	P00233942	M	36-45	3	B	1	0	1	15425

In [46]:
```
## 1.b. Check dataset dimensions
df_walmart.shape
print("Rows and Columns:", df_walmart.shape)
```

```
Rows and Columns: (550068, 10)
```

In [47]:
```
## 1.c. Check column names
df_walmart.columns
```

Out[47]:
```
Index(['User_ID', 'Product_ID', 'Gender', 'Age', 'Occupation', 'City_Category', 'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category',
'Purchase'], dtype='str')
```

In [48]:
```
## 1.d. Check data types and structure
df_walmart.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   User_ID                     550068 non-null  int64
 1   Product_ID                  550068 non-null  str
 2   Gender                      550068 non-null  str
 3   Age                         550068 non-null  str
 4   Occupation                  550068 non-null  int64
 5   City_Category               550068 non-null  str
 6   Stay_In_Current_City_Years  550068 non-null  str
 7   Marital_Status              550068 non-null  int64
 8   Product_Category            550068 non-null  int64
 9   Purchase                    550068 non-null  int64
dtypes: int64(5), str(5)
memory usage: 42.0 MB
```

In [49]:
```
## 1.e. Summary statistics
# For numerical columns:
df_walmart.describe()
```

Out[49]:

	User_ID	Occupation	Marital_Status	Product_Category	Purchase
count	5.500680e+05	550068.000000	550068.000000	550068.000000	550068.000000
mean	1.003029e+06	8.076707	0.409653	5.404270	9263.968713
std	1.727592e+03	6.522660	0.491770	3.936211	5023.065394
min	1.000001e+06	0.000000	0.000000	1.000000	12.000000
25%	1.001516e+06	2.000000	0.000000	1.000000	5823.000000
50%	1.003077e+06	7.000000	0.000000	5.000000	8047.000000
75%	1.004478e+06	14.000000	1.000000	8.000000	12054.000000
max	1.006040e+06	20.000000	1.000000	20.000000	23961.000000

In [50]:
```
## 1.f. Check missing values
df_walmart.isnull().sum()
```

Out[50]:
```
User_ID                     0
Product_ID                  0
Gender                      0
Age                         0
Occupation                  0
City_Category               0
Stay_In_Current_City_Years  0
Marital_Status              0
Product_Category            0
Purchase                    0
dtype: int64
```

In [51]:
```
## 1.g. Check duplicate records
df_walmart.duplicated().sum()
```

Out[51]:
```
np.int64(0)
```

In [52]:
```
## 1.h Analyze categorical variables
categorical_cols = ['Gender','Age','City_Category',
                    'Stay_In_Current_City_Years',
                    'Marital_Status','Product_Category']
for col in categorical_cols:
    print("\n" + "="*40)
    print(col)
    print(df_walmart[col].value_counts())
```

```
========================================
Gender
Gender
M    414259
F    135809
Name: count, dtype: int64

========================================
Age
Age
26-35    219587
36-45    110013
18-25     99660
46-50     45701
51-55     38501
55+       21504
0-17      15102
Name: count, dtype: int64

========================================
City_Category
City_Category
B    231173
C    171175
A    147720
Name: count, dtype: int64

========================================
Stay_In_Current_City_Years
Stay_In_Current_City_Years
1     193821
2     101838
3      95285
4+     84726
0      74398
Name: count, dtype: int64

========================================
Marital_Status
Marital_Status
0    324731
1    225337
Name: count, dtype: int64

========================================
Product_Category
Product_Category
5     150933
1     140378
8     113925
11     24287
2      23864
```

```
6      20466
3      20213
4      11753
16      9828
15      6290
13      5549
10      5125
12      3947
7       3721
18      3125
20      2550
19      1603
14      1523
17       578
9        410
Name: count, dtype: int64
```

In [53]:
```
## 1.i. Check unique values
df_walmart.nunique()
```

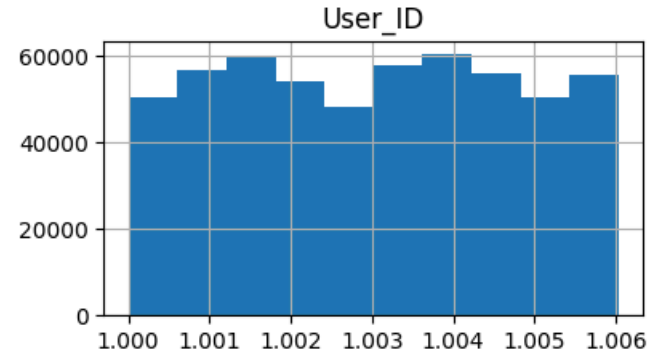
Out[53]:
```
User_ID                     5891
Product_ID                  3631
Gender                         2
Age                            7
Occupation                    21
City_Category                  3
Stay_In_Current_City_Years     5
Marital_Status                 2
Product_Category              20
Purchase                   18105
dtype: int64
```

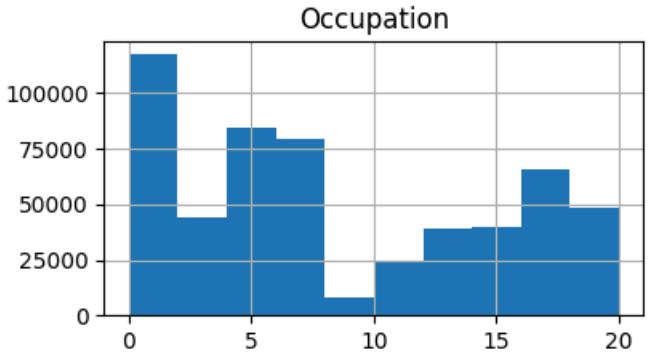
In [54]:
```
## 1.j. Correlation between numerical variables
df_walmart.corr(numeric_only=True)
```

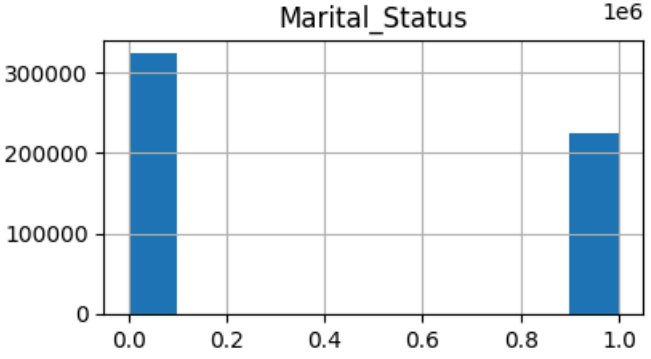
Out[54]:

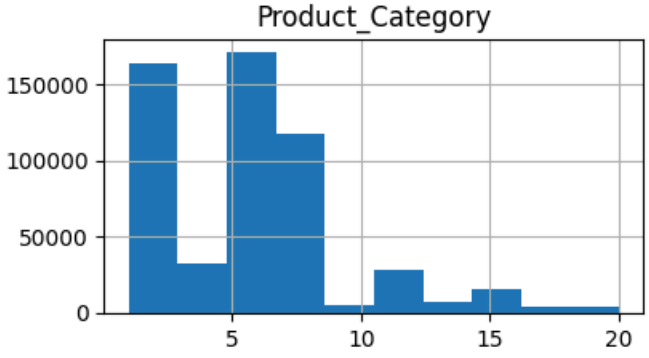
	User_ID	Occupation	Marital_Status	Product_Category	Purchase
User_ID	1.000000	-0.023971	0.020443	0.003825	0.004716
Occupation	-0.023971	1.000000	0.024280	-0.007618	0.020833
Marital_Status	0.020443	0.024280	1.000000	0.019888	-0.000463
Product_Category	0.003825	-0.007618	0.019888	1.000000	-0.343703
Purchase	0.004716	0.020833	-0.000463	-0.343703	1.000000

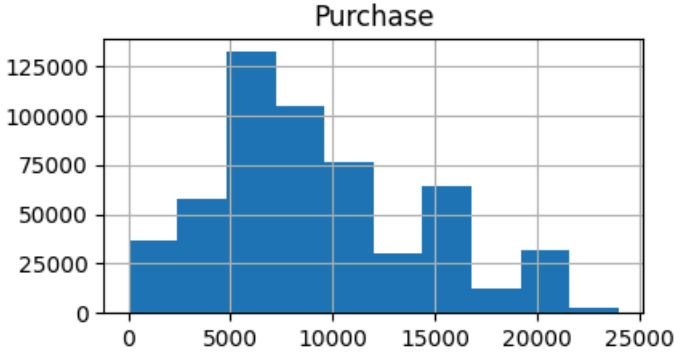
In [55]:
```
## 1.k. Basic visualizations
df_walmart.hist(figsize=(10,8))
plt.show()
```







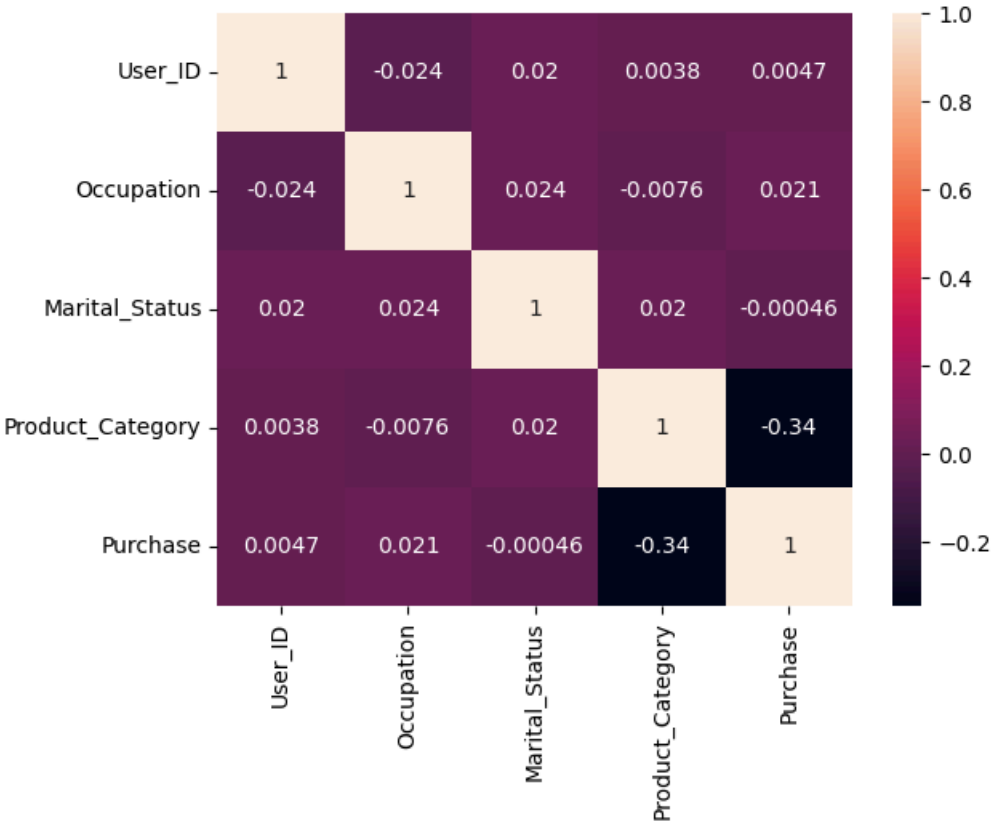




In [56]:
```
## Observations from Histograms :
# Purchase
# The Purchase variable is widely distributed, indicating substantial variation in customer spending.
# Most transactions are concentrated around the middle purchase range, while a smaller number of transactions have very high purchase values.
# The distribution appears slightly skewed, suggesting the presence of high-spending customers.
# Occupation
# Customer transactions are distributed across multiple occupations.
# Some occupations contribute significantly more transactions than others, indicating differences in customer representation.
# Product Category
# Purchases are concentrated in a few product categories.
# Certain categories are considerably more popular among customers.
```

```
# Marital Status
# Both married and unmarried customers are well represented in the dataset.
# The distribution appears relatively balanced, with unmarried customers slightly more frequent.
# Overall Observation
# The dataset exhibits diverse customer characteristics and purchasing behaviors.
# No unusual or impossible values are observed from the histogram distributions.
```

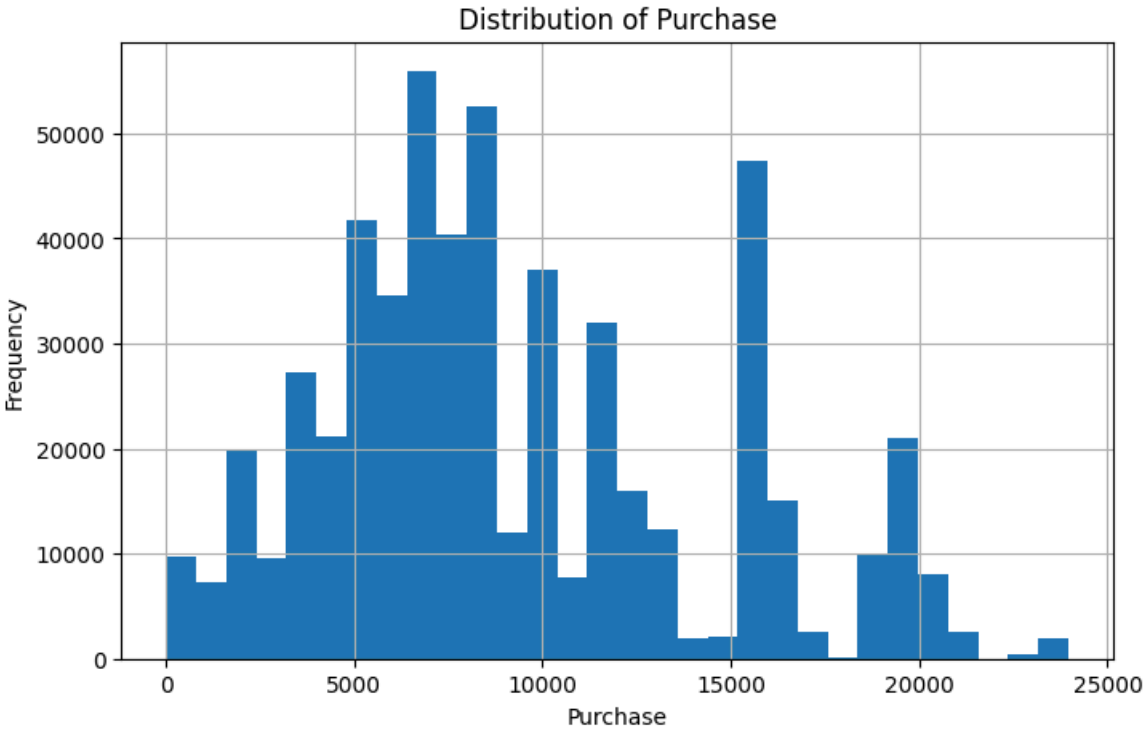
In [57]:
```
## 1.l.
sns.heatmap(df_walmart.corr(numeric_only=True), annot=True)
plt.show()
```



In [58]:
```
## Observations from the Correlation Heatmap:
# Most variables show weak correlations with each other, indicating that customer purchasing behavior is influenced by multiple factors
# rather than a single attribute.
# Purchase has a very low correlation with Occupation and Marital_Status, suggesting that these variables alone do not strongly
# impact the purchase amount.
# Product_Category shows a weak negative correlation with Purchase, indicating that spending varies across product categories
# but the relationship is not strong.
# The absence of strong positive or negative correlations suggests that there is no multicollinearity issue among the numerical variables.
```

```
# Overall, the heatmap indicates that demographic attributes such as occupation and marital status have limited linear relationships
# with purchase amount, highlighting the need for deeper segmentation analysis using variables like gender and age.
```

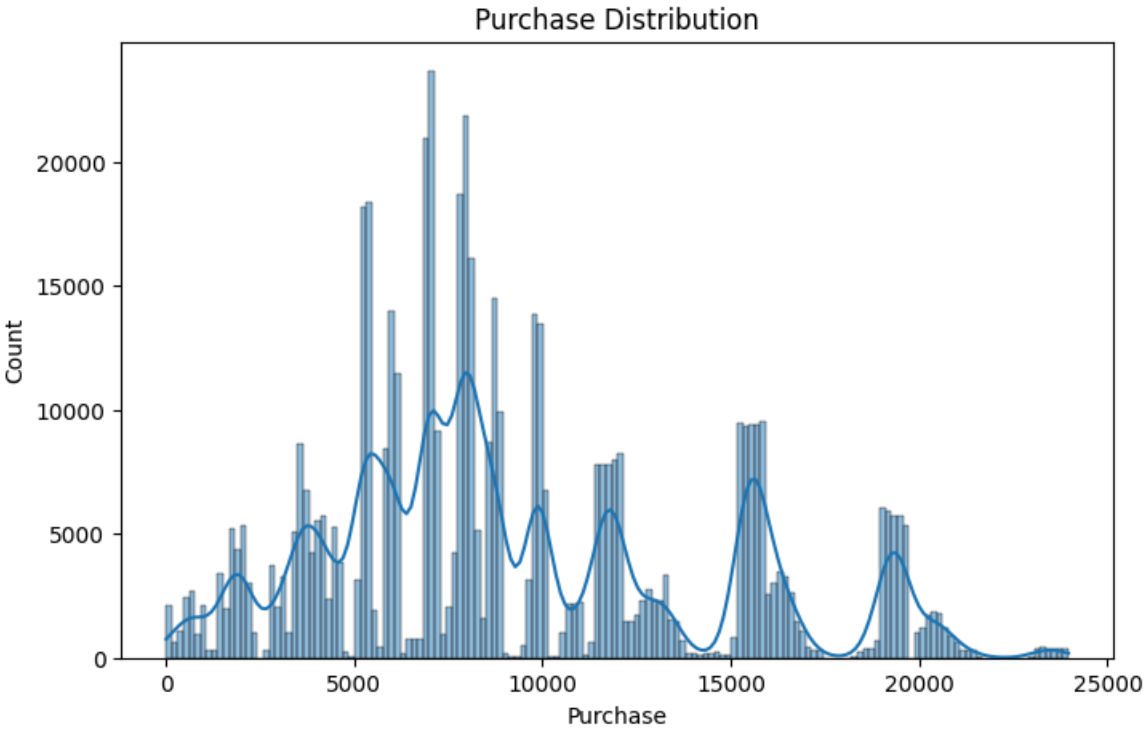
In [59]:
```
## 1.m. Univariate Analysis
# Histogram
plt.figure(figsize=(8,5))
df_walmart['Purchase'].hist(bins=30)
plt.title('Distribution of Purchase')
plt.xlabel('Purchase')
plt.ylabel('Frequency')
plt.show()
```



In [60]:
```
## Observations from Purchase Distribution Histogram:
# The distribution of Purchase Amount is not perfectly symmetric and exhibits a slight skew,
# indicating that customer spending is not evenly distributed.
# Most transactions are concentrated in the mid-range purchase values, suggesting that the majority of customers make moderate purchases.
# A smaller number of transactions occur at very high purchase amounts, indicating the presence of high-spending customers.
# The wide spread of the distribution reflects significant variability in customer purchasing behavior.
# The absence of unusual gaps or spikes suggests that the purchase data is reasonably continuous and suitable for further statistical analysis.
```

In [61]:
```
# Distplot / KDE Plot
plt.figure(figsize=(8,5))
sns.histplot(df_walmart['Purchase'], kde=True)
```

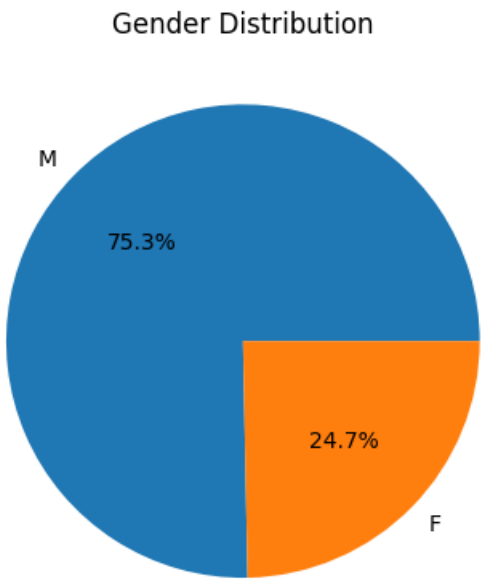
```
plt.title('Purchase Distribution')
plt.show()
```



In [62]:
```
# Observations from Purchase Distribution (Histogram with KDE)
# The distribution of Purchase Amount is slightly skewed, indicating that customer spending is not normally distributed.
# The histogram shows that a large number of transactions are concentrated around the middle purchase range,
# while fewer transactions occur at extremely low or high purchase values.
# The KDE (Kernel Density Estimate) curve helps visualize the overall shape of the distribution and confirms the presence of one major peak,
# suggesting a dominant spending range among customers.
# The long tail toward higher purchase amounts indicates the presence of customers who make relatively large purchases.
# The distribution exhibits substantial variability in spending behavior, reflecting differences in customer preferences,
# demographics, and purchasing power.
```

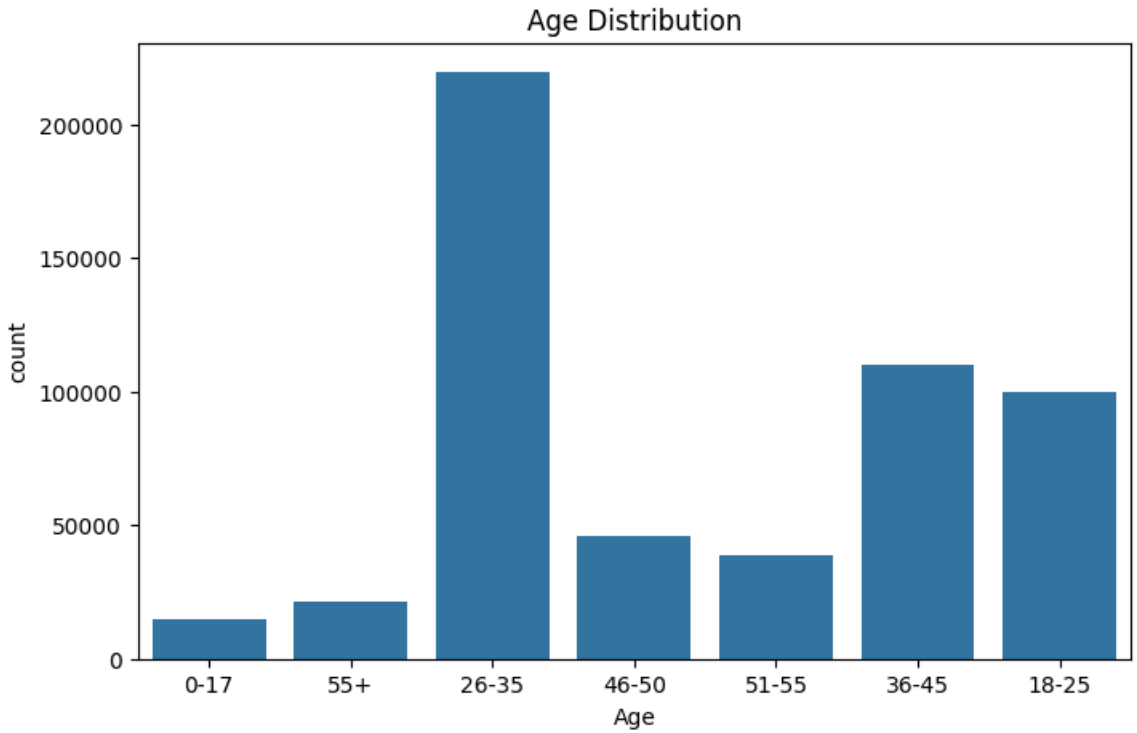
In [63]:
```
## Univariate Analysis of Categorical Variables
# Gender
df_walmart['Gender'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title('Gender Distribution')
plt.ylabel('')
plt.show()
```



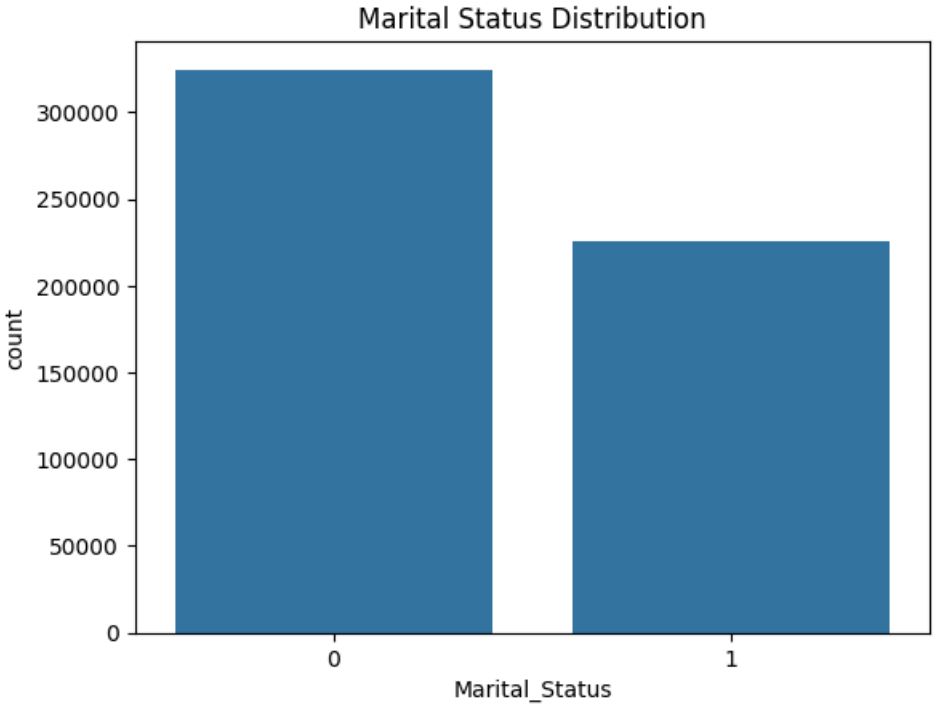
In [64]:
```
## Observations from Gender Distribution :
# Male customers constitute a larger share of the dataset than female customers.
# Both genders are well represented in the customer base.
# The dataset is slightly male-dominated, which may influence overall purchasing trends.
```

In [65]:
```
# Age
plt.figure(figsize=(8,5))
sns.countplot(data=df_walmart, x='Age')
plt.title('Age Distribution')
plt.show()
```



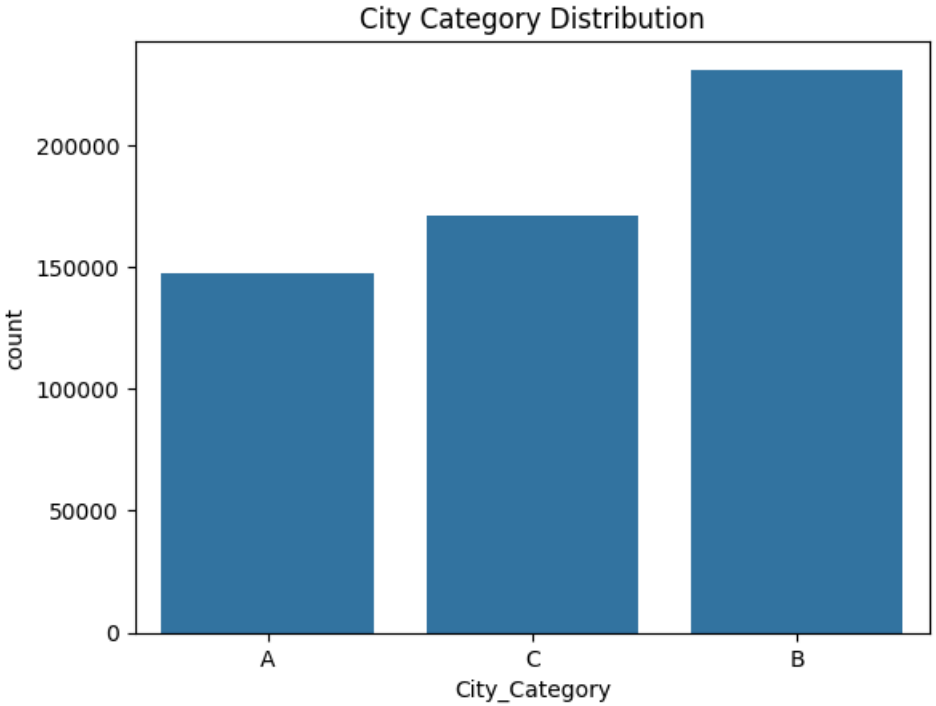
In [ ]:

In [66]:
```
# Marital Status
sns.countplot(data=df_walmart, x='Marital_Status')
plt.title('Marital Status Distribution')
plt.show()
```



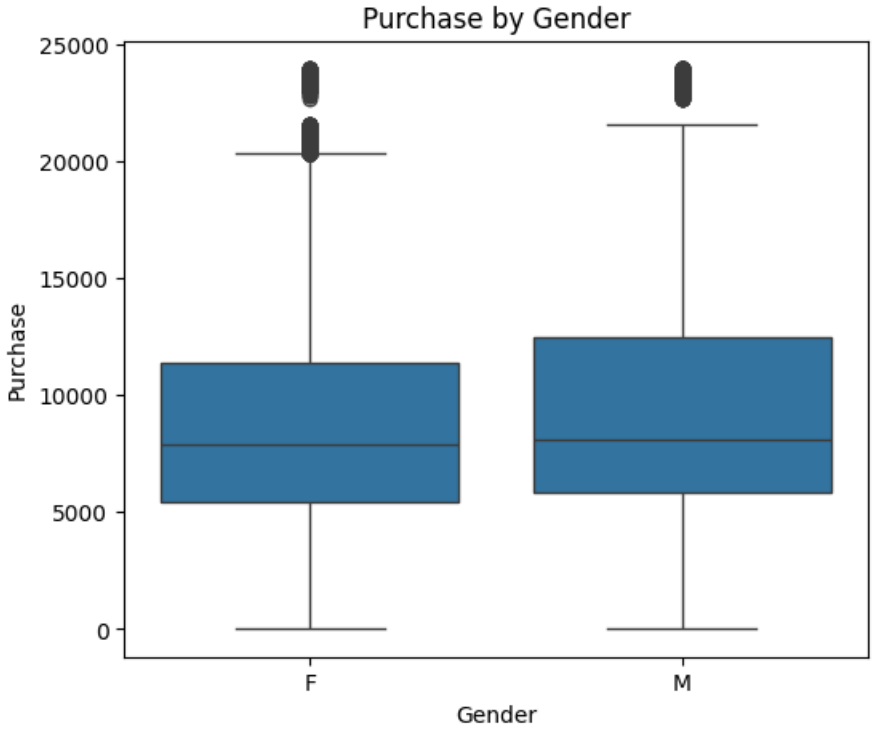
In [67]:
```
## Observations from Age Distribution :
# The 26-35 age group has the highest number of customers.
# The majority of customers belong to the 18-50 years age range.
# The 0-17 and 51+ age groups have comparatively fewer customers.
# Walmart's customer base is primarily composed of working-age adults.
```

In [68]:
```
# City Category
sns.countplot(data=df_walmart, x='City_Category')
plt.title('City Category Distribution')
plt.show()
```



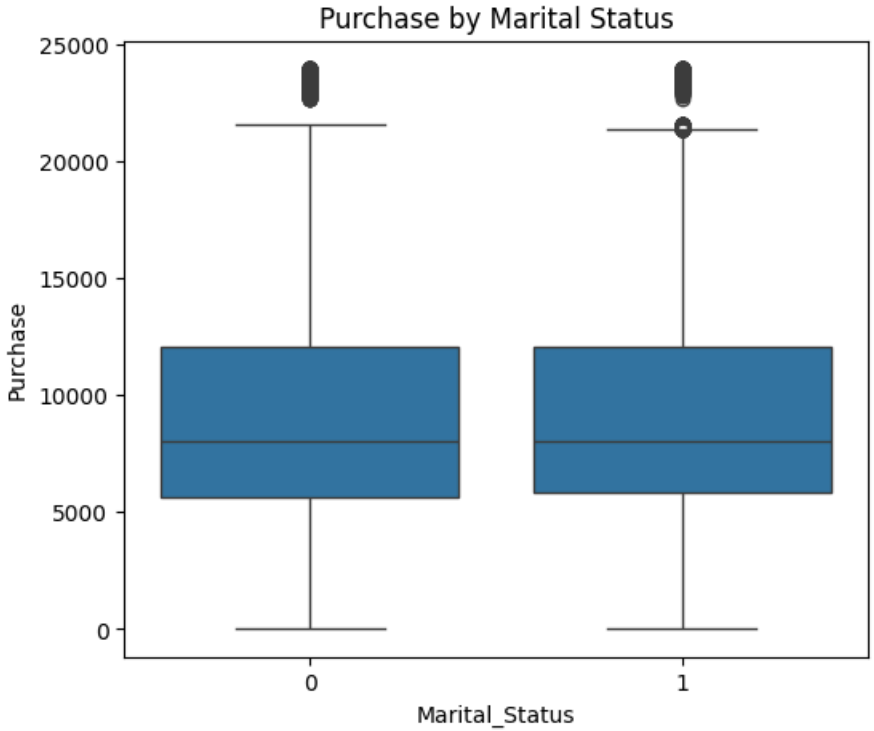
In [69]:
```
## Observations from Marital Status Distribution
# Unmarried customers slightly outnumber married customers.
# Both married and unmarried customers are well represented in the dataset.
# The distribution is relatively balanced between the two groups.
```

In [70]:
```
# Bivariate Analysis (Categorical vs Continuous)
# Gender vs Purchase
plt.figure(figsize=(6,5))
sns.boxplot(data=df_walmart, x='Gender', y='Purchase')
plt.title('Purchase by Gender')
plt.show()
```



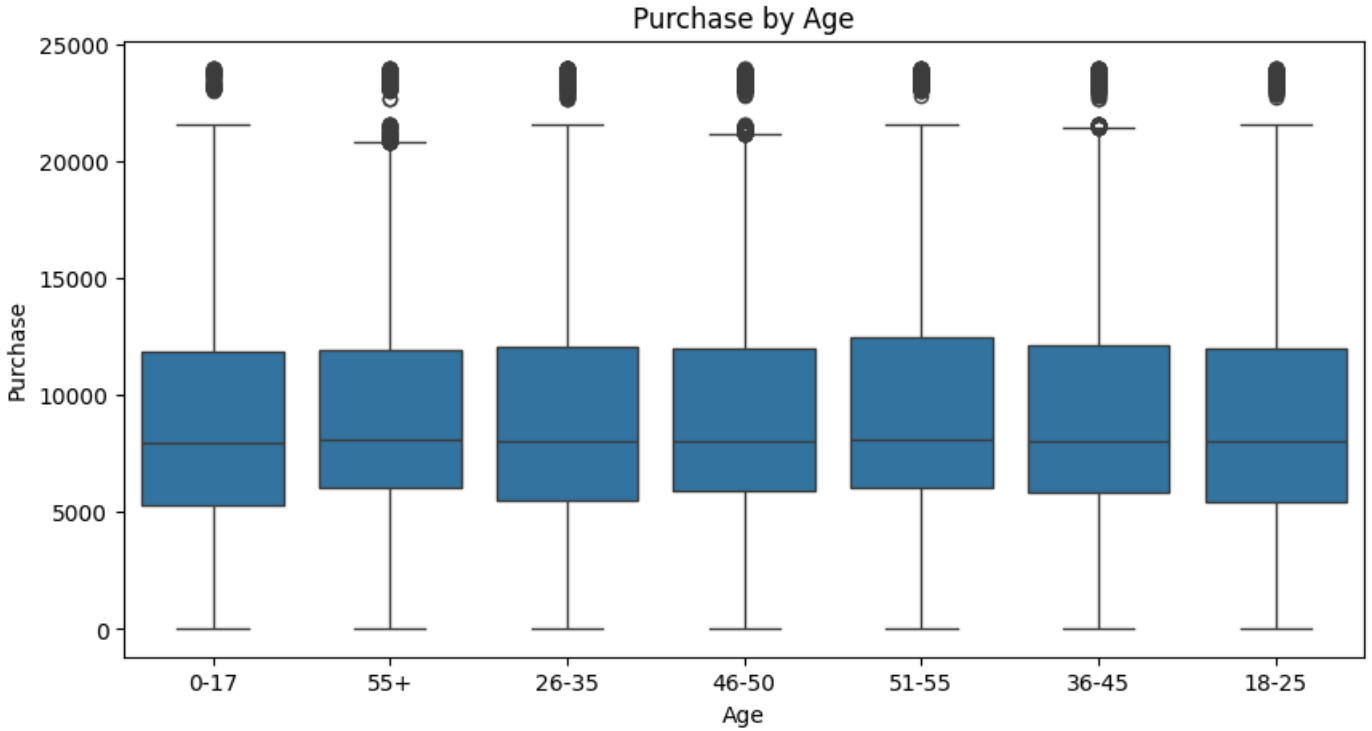
In [71]:
```
## Observation:
# Male customers tend to have a higher median purchase amount than females.
```

In [72]:
```
# Marital Status vs Purchase
plt.figure(figsize=(6,5))
sns.boxplot(data=df_walmart, x='Marital_Status', y='Purchase')
plt.title('Purchase by Marital Status')
plt.show()
```



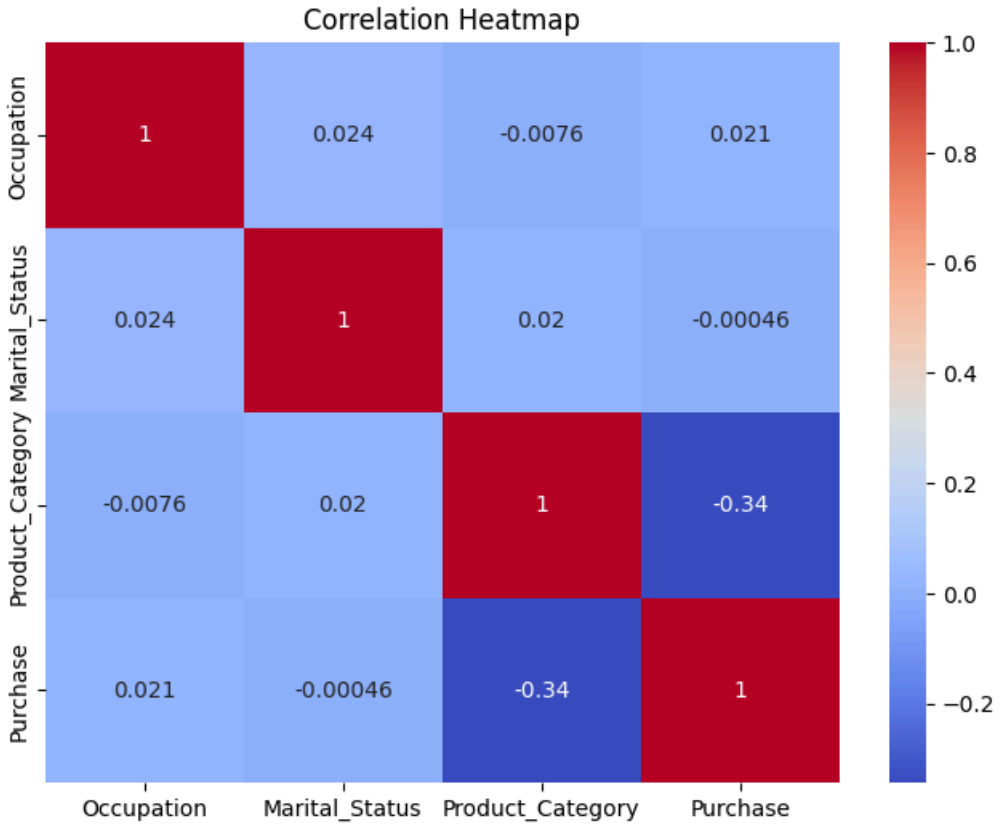
In [73]:
```
## Observation:
# Married and unmarried customers show similar spending distributions.
```

In [74]:
```
# Age vs Purchase
plt.figure(figsize=(10,5))
sns.boxplot(data=df_walmart, x='Age', y='Purchase')
plt.title('Purchase by Age')
plt.show()
```



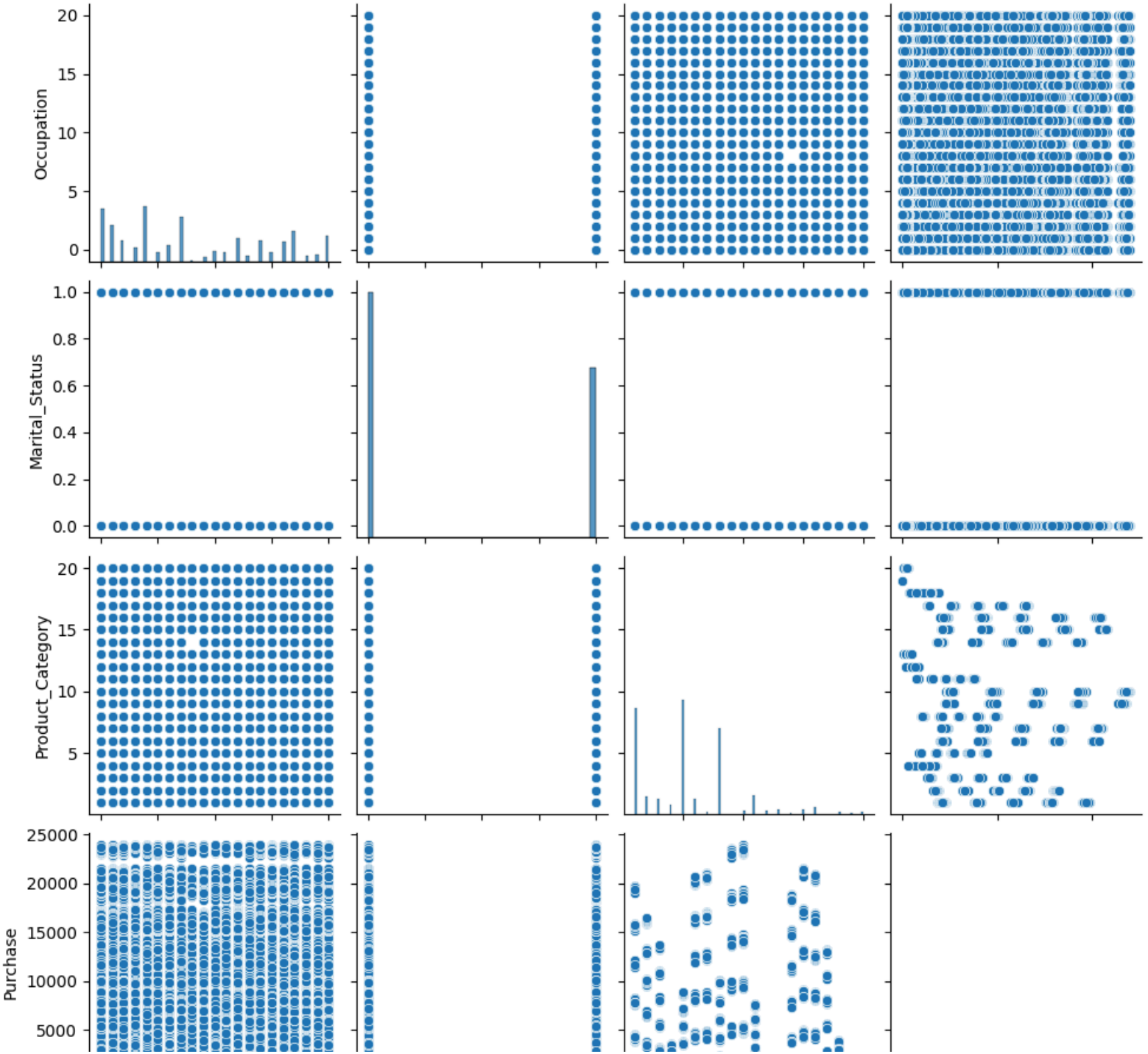
In [75]:
```
## Observation :
# Median purchase amount generally increases with age.
# Older age groups show higher spending behavior.
```

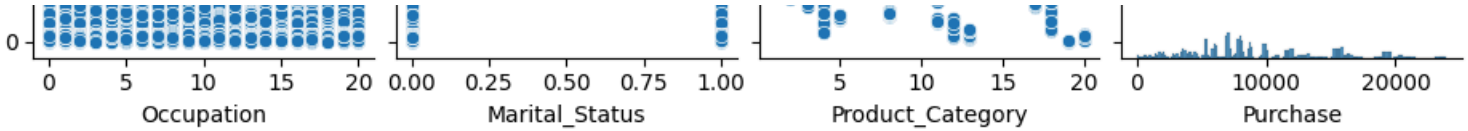
In [76]:
```
# Correlation Analysis
# Since most variables are categorical, create a correlation matrix using numerical columns.
plt.figure(figsize=(8,6))
corr = df_walmart[['Occupation',
                   'Marital_Status',
                   'Product_Category',
                   'Purchase']].corr()
sns.heatmap(corr,
            annot=True,
            cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



In [77]:
```
## Observations :
# Most correlations are weak.
# Purchase amount is not strongly correlated with any single numerical variable.
# Customer spending is likely influenced by multiple factors together.
```

In [78]:
```
# Pairplot
sns.pairplot(
    df_walmart[['Occupation',
                'Marital_Status',
                'Product_Category',
                'Purchase']]
)
plt.show()
```


Occupation
Marital_Status
Product_Category
Purchase
20
15
10
5
0
1.0
0.8
0.6
0.4
0.2
0.0
25000
20000
15000
10000
5000



In [79]:
```
## Observations :
# No strong linear relationships are visible.
# Purchase values are widely dispersed across occupations and product categories.
# This suggests customer spending behavior is complex and not driven by a single variable.
```

In [80]:
```
# Insights :
# Dataset contains 550068 rows and 10 columns.
# Numerical columns: User_ID, Occupation,Marital_Status, Product_Category, Purchase.
# Categorical columns: Gender,Age,City_Category,Stay_In_Current_City_Years,Marital_Status,Product_Category.
# There are no missing values in the given dataset.
# There are no duplicate records in the given dataset.
# The average purchase amount is 9263.96(9263.968713), while the median is 8,047. Since the mean is greater than the median,
# the distribution is positively skewed.
# The standard deviation of 5023(5023.065394) indicates substantial variation in purchase amounts. Purchase values range from 12 to 23,961.
# The Purchase variable is concentrated around the median value of 8,047 (or mean value of 9,263) and shows a right-skewed distribution,
# as the mean is higher than the median. Most purchase amounts lie between 5,823 and 12,054 (the interquartile range).
# Purchase has a weak positive correlation with Occupation (0.12), indicating a slight increase in purchase amount across occupation categories.
# Purchase shows almost no correlation with Marital_Status (0.01). Product_Category has a moderate negative correlation (-0.34) with Purchase.
# A correlation analysis was performed on the numerical columns.
# Most correlations were weak, indicating limited linear relationships among the variables.
# Since several variables (Occupation, Marital_Status, and Product_Category) are categorical codes rather than true continuous measures,
# correlation values should be interpreted with caution.
```

In [81]:
```
# 2. Detect Null values & Outliers (using boxplot, “describe” method by checking the difference between mean and median, isnull etc.)
# 2.a. Null Value Analysis
df_walmart.isnull().sum()
```

Out[81]:
```
User_ID                     0
Product_ID                  0
Gender                      0
Age                         0
Occupation                  0
City_Category               0
Stay_In_Current_City_Years  0
Marital_Status              0
Product_Category            0
Purchase                    0
dtype: int64
```

In [82]:
```
# Observation:
# No missing values were found in any of the columns.
# The dataset contains 550,068 records and is complete, making it suitable for further analysis without the need for data imputation or
# removal of records.
```

In [83]:
```
## Insight:
# Since there are no null values, the quality of the dataset is good and the analysis results are less likely to be affected by missing information.
```

In [84]:
```
plt.figure(figsize=(8,5))
plt.boxplot(df_walmart['Purchase'])
plt.title('Boxplot of Purchase Amount')
plt.ylabel('Purchase')
plt.show()
```



In [85]:
```
# Observation:
# The boxplot displays observations beyond the upper whisker, indicating potential outliers in the Purchase variable.
# Most purchase amounts are concentrated within the interquartile range (₹5,823 - ₹12,054).
```

In [86]:
```
## Insight:
# The outliers represent customers with unusually high purchase amounts. These transactions may correspond to premium product purchases or
# bulk purchases and can significantly influence measures such as the mean.
# Therefore, median and percentile-based measures provide a more robust representation of typical customer spending behavior.
```

In [87]:
```
## Conclusion
# The Walmart dataset does not contain any missing values. Outlier analysis using descriptive statistics and boxplots indicates that the
# Purchase variable is right-skewed and contains high-value transactions.
# While these observations may be treated as outliers from a statistical perspective,
# they can also provide valuable business insights by identifying customers who contribute significantly to revenue.
```

In [88]:
```
# 3. Do some data exploration steps like:
# 3.a. Tracking the amount spent per transaction of all the 50 million female customers, and all the 50 million male customers,
# calculate the average, and conclude the results.

df_walmart.groupby('Gender')['Purchase'].agg(['count','mean','median','std'])
```

Out[88]:

Gender	count	mean	median	std
F	135809	8734.565765	7914.0	4767.233289
M	414259	9437.526040	8098.0	5092.186210

In [89]:
```
# Male customers recorded an average purchase amount of 99437(9437.526040), while female customers recorded an average purchase amount of
# 8734(8734.565765).
# The median purchase values and standard deviations further indicate the spending pattern and variability within each gender group.
# Based on the average transaction value, male customers tend to spend more per transaction than female customers.
```

In [90]:
```
# 3.b. Inference after computing the average female and male expenses.
df_walmart.groupby('Gender')['Purchase'].mean()
```

Out[90]:
```
Gender
F    8734.565765
M    9437.526040
Name: Purchase, dtype: float64
```

In [91]:
```
# Inference:
# Male customers have a higher average purchase amount than female customers. On average, male customers spend more per transaction at Walmart.
# This suggests that gender may influence purchasing behavior, with male customers contributing higher transaction values.
```

In [92]:
```
# 3.c. Use the sample average to find out an interval within which the population average will lie.
# Using the sample of female customers you will calculate the interval within which the average spending of
# 50 million male and female customers may lie.
female = df_walmart[df_walmart['Gender'] == 'F']['Purchase']

female_mean = female.mean()
female_std = female.std()
female_n = female.count()

# Calculate a 95% Confidence Interval
margin_error = 1.96 * (female_std / np.sqrt(female_n))

lower = female_mean - margin_error
upper = female_mean + margin_error

print("95% Confidence Interval:")
print(lower, upper)
```

```
95% Confidence Interval:
8709.211081242413 8759.920449068539
```

In [93]:
```
# The sample average purchase amount for female customers is an estimate of the population average.
# Based on the confidence interval, the average spending of all female Walmart customers is likely to be between 8,709.21 and 8,759.92.
```

In [94]:
```
## Insight
# The confidence interval is relatively narrow, indicating that the sample provides a reliable estimate of the population's
# average spending behavior.
# Walmart can use this estimate to better understand the purchasing patterns of female customers and design targeted marketing strategies.
```

In [95]:
```
male = df_walmart[df_walmart['Gender'] == 'M']['Purchase']

male_mean = male.mean()
male_std = male.std()
male_n = male.count()

# Calculate a 95% Confidence Interval
margin_error = 1.96 * (male_std / np.sqrt(male_n))

lower = male_mean - margin_error
upper = male_mean + margin_error

print("95% Confidence Interval:")
print(lower, upper)
```

```
95% Confidence Interval:
9422.019162420047 9453.032918524483
```

In [96]:
```
## Insights
# Male customers exhibit a higher average spending pattern compared to female customers.
# Walmart may use this information to design gender-specific marketing campaigns, promotional offers, and product recommendations.
# The narrow confidence intervals indicate that the estimates are reliable and the sample size is sufficiently large to represent the customer
# population accurately.
```

In [97]:
```
# 4. Use the Central limit theorem to compute the interval. Change the sample size to observe the distribution of the mean of the expenses by female
# and male customers.
# The interval that you calculated is called Confidence Interval. The width of the interval is mostly decided by the business:
# Typically 90%, 95%, or 99%. Play around with the width parameter and report the observations.
print(female.head())
print(male.head())
```

```
0      8370
1     15200
2      1422
3      1057
14     5378
Name: Purchase, dtype: int64
4     7969
5    15227
6    19215
7    15854
8    15686
Name: Purchase, dtype: int64
```

In [98]:
```python
# Confidence Interval Calculation
def confidence_interval(data, confidence=0.95):
    mean = data.mean()
    std = data.std()

    n = len(data)

    z = norm.ppf((1 + confidence)/2)

    margin_error = z * (std/np.sqrt(n))

    lower = mean - margin_error
    upper = mean + margin_error

    return lower, upper
```

In [99]:
```python
# For Female
for conf in [0.90, 0.95, 0.99]:
    lower, upper = confidence_interval(female, conf)
    print(f"{int(conf*100)}% CI : ({lower:.2f}, {upper:.2f})")
```

```
90% CI : (8713.29, 8755.84)
95% CI : (8709.21, 8759.92)
99% CI : (8701.24, 8767.89)
```

In [100…
```python
# For Male
for conf in [0.90, 0.95, 0.99]:
    lower, upper = confidence_interval(male, conf)
    print(f"{int(conf*100)}% CI : ({lower:.2f}, {upper:.2f})")
```

```
90% CI : (9424.51, 9450.54)
95% CI : (9422.02, 9453.03)
99% CI : (9417.15, 9457.91)
```

In [101…
```python
for conf in [0.90, 0.95, 0.99]:
    lower, upper = confidence_interval(female, conf)
    width = upper - lower
    print(f"{int(conf*100)}% CI : ({lower:.2f}, {upper:.2f})")
    print(f"Width : {width:.2f}")
    print("-"*40)
```

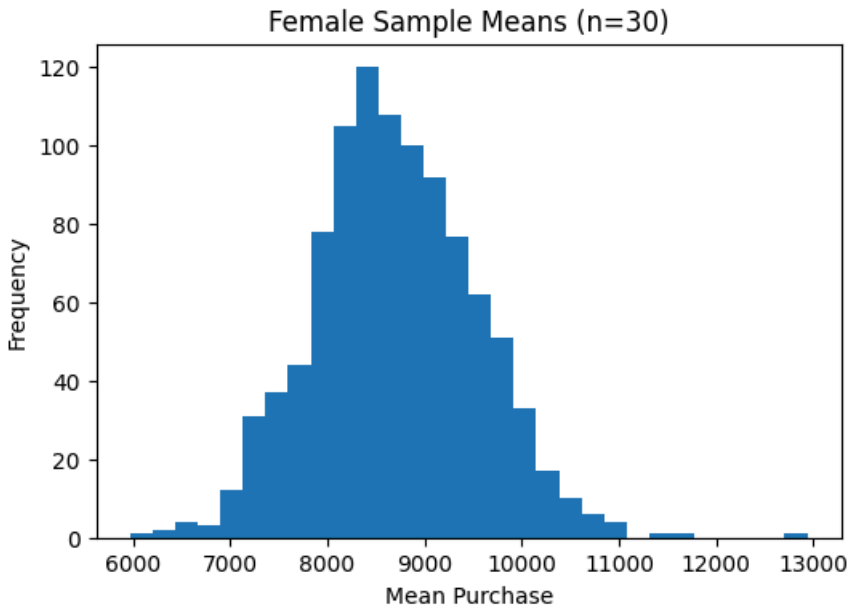
```
90% CI : (8713.29, 8755.84)
Width : 42.56
----------------------------------------
95% CI : (8709.21, 8759.92)
Width : 50.71
----------------------------------------
99% CI : (8701.24, 8767.89)
Width : 66.64
----------------------------------------
```

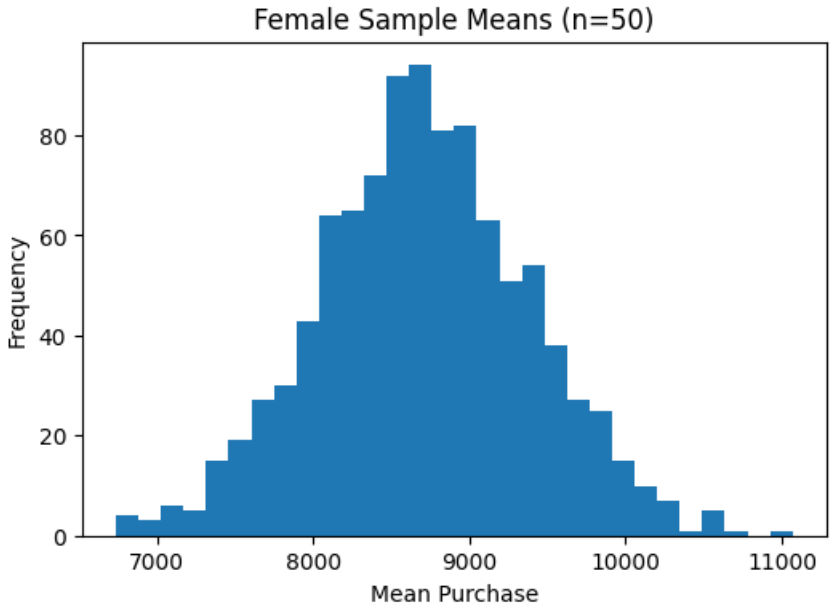
In [102…
```python
# CLT Visualization
```

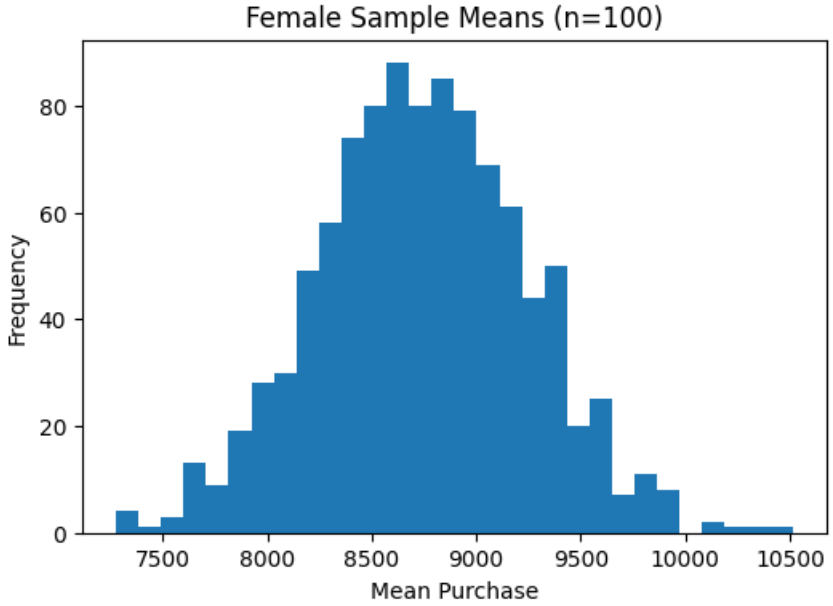
```
sample_sizes = [30, 50, 100, 500]
```

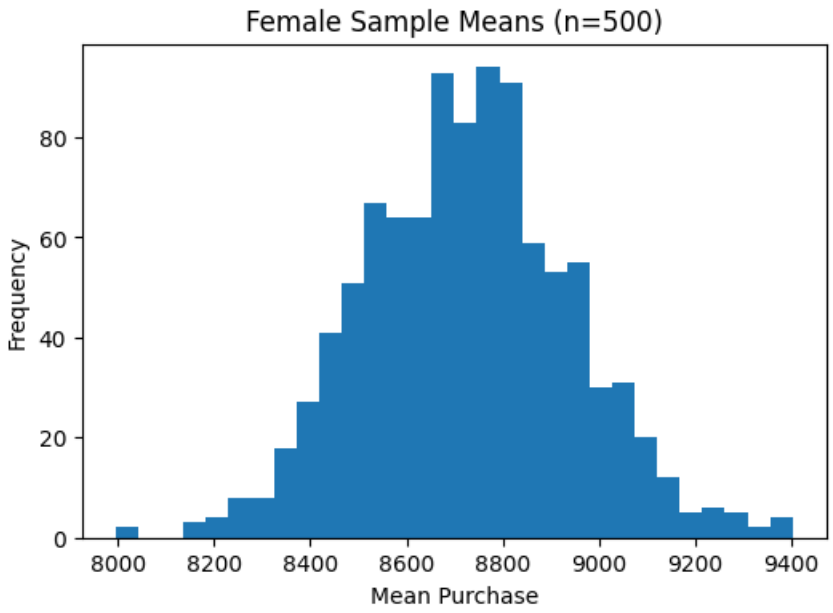
In [103…

```
# For Female
for n in sample_sizes:
    sample_means = []
    for i in range(1000):
        sample = female.sample(n, replace=True)
        sample_means.append(sample.mean())
    plt.figure(figsize=(6,4))
    plt.hist(sample_means, bins=30)
    plt.title(f'Female Sample Means (n={n})')
    plt.xlabel('Mean Purchase')
    plt.ylabel('Frequency')
    plt.show()
```




Female Sample Means (n=50)
Frequency
0
20
40
60
80
7000
8000
9000
10000
11000
Mean Purchase


Female Sample Means (n=100)
Frequency
0
20
40
60
80
7500
8000
8500
9000
9500
10000
10500
Mean Purchase

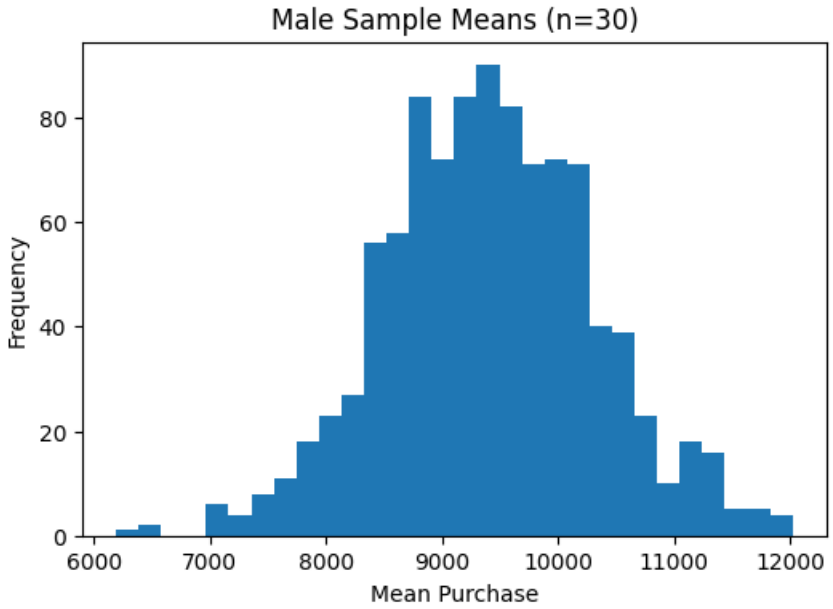


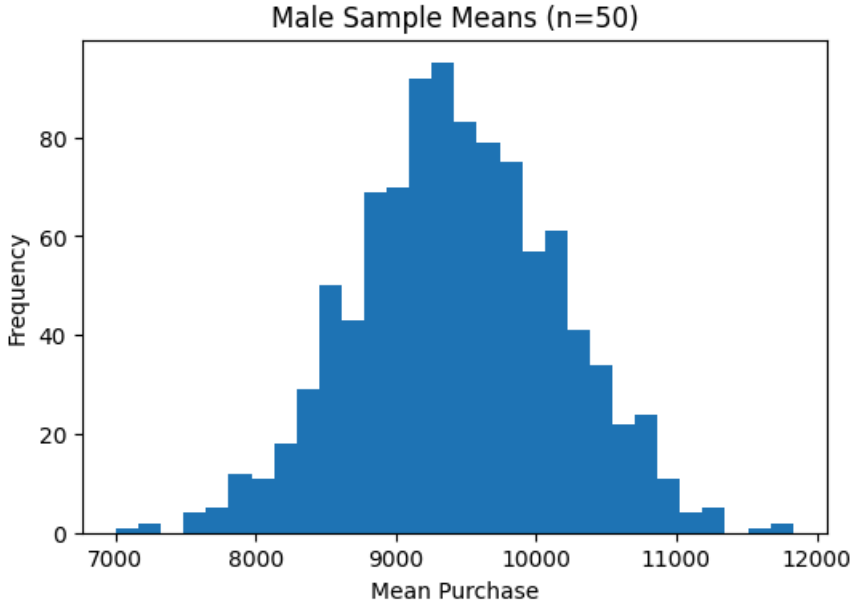
In [104…

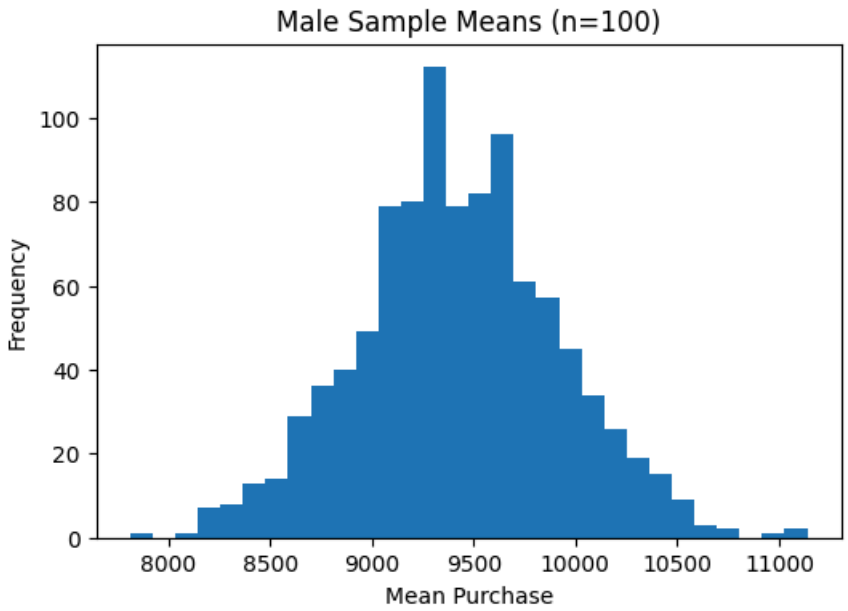
```
## Insights :
# The average purchase amount for female customers is estimated to lie between ₹8709.21 and ₹8759.92 at a 95% confidence level.
# As the confidence level increases from 90% → 95% → 99%, the confidence interval becomes wider:
# Width at 90% = 42.55
# Width at 95% = 50.71
# Width at 99% = 66.65
# The wider interval at higher confidence levels indicates greater certainty that the true population mean spending of female customers
# lies within the interval.
# The increase in interval width demonstrates the trade-off between:
# Precision (narrower intervals at lower confidence levels)
# Confidence (greater certainty at higher confidence levels)
# The confidence intervals are wider for female customers than for male customers, suggesting slightly greater variability in
# female customer spending.
# The estimated average spending for female customers is consistently lower than that of male customers, indicating that
# female customers spend less on average at Walmart.
```

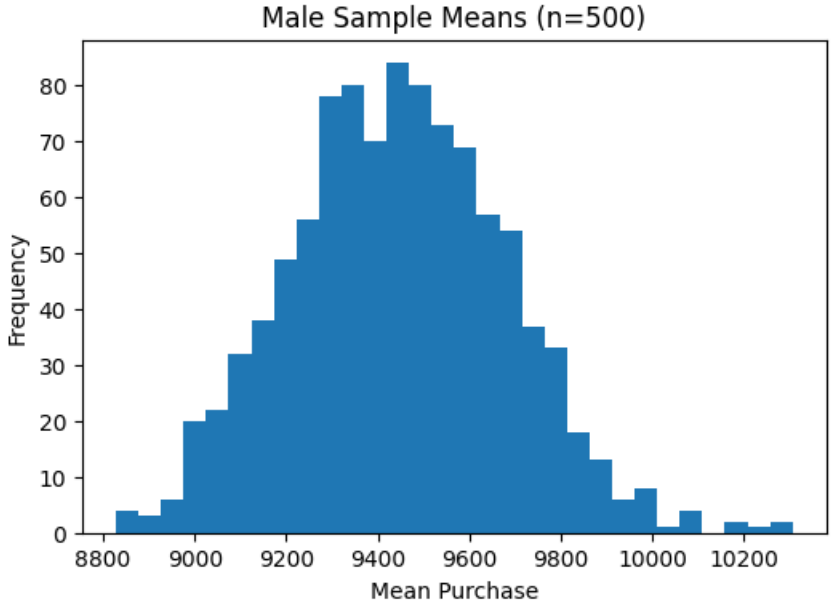
In [105…

```
# For Male
for n in sample_sizes:
    sample_means = []
    for i in range(1000):
        sample = male.sample(n, replace=True)
        sample_means.append(sample.mean())
    plt.figure(figsize=(6,4))
    plt.hist(sample_means, bins=30)
    plt.title(f'Male Sample Means (n={n})')
    plt.xlabel('Mean Purchase')
    plt.ylabel('Frequency')
    plt.show()
```


Male Sample Means (n=30)
Frequency
0
20
40
60
80
6000
7000
8000
9000
10000
11000
12000
Mean Purchase


Male Sample Means (n=50)
Frequency
0
20
40
60
80
7000
8000
9000
10000
11000
12000
Mean Purchase





In [106…

```
## Insights :
# The average purchase amount for male customers is estimated to lie between ₹9422.02 and ₹9453.03 at a 95% confidence level.
# As the confidence level increases from 90% → 95% → 99%, the confidence interval becomes wider:
# Width at 90% = 26.03
# Width at 95% = 31.01
```

```
# Width at 99% = 40.76
# The wider interval at higher confidence levels indicates a greater certainty that the true population mean spending lies within the interval.
# The increase in interval width demonstrates the trade-off between:
# Precision (narrower intervals)
# Confidence (greater certainty)
# The confidence intervals for male customers are relatively narrow, suggesting a stable estimate of the average spending behavior
# of male customers.
# Comparing with female customers, the entire male confidence interval lies above the female confidence interval,
# indicating that male customers spend
# more on average than female customers at Walmart.
```

In [107…

```
# # Combined Analysis :
# As confidence level increases from 90% to 99%, the confidence intervals become wider for both genders.
# Wider intervals indicate greater confidence that the true population mean lies within the interval.
# Male customers have a higher average purchase amount than female customers because all male confidence intervals lie above the
# corresponding female confidence intervals.
# Male confidence intervals are narrower than female confidence intervals, suggesting a more precise estimate of the male average purchase amount.
```

In [108…

```
# 5. Conclude the results and check if the confidence intervals of average male and female spends are overlapping or not overlapping.
# How can Walmart leverage this conclusion to make changes or improvements?
female_ci = (8709.21, 8759.92)
male_ci = (9422.02, 9453.03)

if female_ci[1] < male_ci[0] or male_ci[1] < female_ci[0]:
    print("The confidence intervals do not overlap.")
else:
    print("The confidence intervals overlap.")
```

```
The confidence intervals do not overlap.
```

In [109…

```
## Interpretation.
# The confidence intervals of male and female spending are completely separate.
# This indicates a significant difference in the average spending behavior of male and female customers.
# Male customers spend considerably more on average than female customers.
# The difference is unlikely to be due to random sampling variation.
# Gender appears to be an important factor influencing purchase amount.
```

In [110…

```
## Business Insights :
## Implement Gender-Specific Marketing Campaigns
#  Since male customers spend more on average, Walmart can design targeted promotions and advertisements based on
#  gender-specific purchasing behavior.
#  Personalized marketing can improve customer engagement and conversion rates.
## Optimize Product Assortment
#  Walmart can analyze which product categories contribute most to male spending and ensure adequate inventory in those categories.
#  Similarly, products preferred by female customers can be bundled or promoted to encourage higher spending.
## Introduce Personalized Offers
#  Female customers can be targeted with discounts, cashback offers, and bundle deals to increase their average purchase value.
#  Male customers can be offered premium product recommendations and loyalty rewards to maintain their spending levels.
## Enhance Customer Segmentation
#  The spending difference suggests that gender is an important customer segment.
#  Walmart can create separate customer profiles and marketing strategies for different demographic groups.
## Improve Revenue Through Cross-Selling
```

```
#  Walmart can recommend complementary products based on gender-specific shopping patterns.
#  This can increase basket size and overall sales.
## Design Better Loyalty Programs
#  Special rewards and incentives can be introduced for customer groups with lower average spending to encourage repeat purchases and higher transacti
#  Data-Driven Decision Making
#  Since the confidence intervals do not overlap, the difference in spending behavior is statistically meaningful.
#  Walmart can confidently use this insight when making merchandising, pricing, and promotional decisions.

## Overall :
# Since the male and female confidence intervals do not overlap, there is strong evidence that male customers have a higher average purchase amount
# than female customers, suggesting that Walmart should adopt gender-specific marketing and promotional strategies.
```

In [111…

```
##6. Perform the same activity for Married vs Unmarried and Age
#    For Age, you can try bins based on life stages: 0-17, 18-25, 26-35, 36-50, 51+ years.

# Married vs Unmarried Datasets
married = df_walmart[df_walmart['Marital_Status'] == 1]['Purchase']
unmarried = df_walmart[df_walmart['Marital_Status'] == 0]['Purchase']

print("Married customers:", len(married))
print("Unmarried customers:", len(unmarried))
```

```
Married customers: 225337
Unmarried customers: 324731
```

In [112…

```
# Confidence Interval Function
def confidence_interval(data, confidence=0.95):

    mean = data.mean()
    std = data.std()
    n = len(data)

    z = norm.ppf((1 + confidence)/2)

    margin_error = z * (std/np.sqrt(n))

    lower = mean - margin_error
    upper = mean + margin_error

    return lower, upper
```

In [113…

```
# Calculate CIs for Married
for conf in [0.90, 0.95, 0.99]:
    lower, upper = confidence_interval(married, conf)
    print(f"Married {int(conf*100)}% CI : ({lower:.2f}, {upper:.2f})")
```

```
Married 90% CI : (9243.79, 9278.56)
Married 95% CI : (9240.46, 9281.89)
Married 99% CI : (9233.95, 9288.40)
```

In [114…

```
## Calculate CIs for Married
for conf in [0.90, 0.95, 0.99]:
```

```
    lower, upper = confidence_interval(unmarried, conf)
    print(f"Unmarried {int(conf*100)}% CI : ({lower:.2f}, {upper:.2f})")
```

```
Unmarried 90% CI : (9251.40, 9280.42)
Unmarried 95% CI : (9248.62, 9283.20)
Unmarried 99% CI : (9243.18, 9288.63)
```

In [115…

```
# Check Overlap for both cases.
married_ci = confidence_interval(married,0.95)
unmarried_ci = confidence_interval(unmarried,0.95)

if married_ci[1] < unmarried_ci[0] or unmarried_ci[1] < married_ci[0]:
    print("Confidence intervals do not overlap")
else:
    print("Confidence intervals overlap")
```

```
Confidence intervals overlap
```

In [116…

```
## Business Recommendations for Married vs Unmarried Customers :
# Concrete Recommendations
# 1. Same Pricing Strategy for Both Groups
# Since spending behavior is similar, Walmart does not need separate pricing structures for married and unmarried customers.
# A common pricing and promotion strategy can be applied to both groups.
# 2. Focus Marketing Budget Elsewhere
# The gender analysis showed a much larger difference than marital status.
# Walmart should invest more marketing resources in gender-based and age-based campaigns, where spending differences are more pronounced.
# 3. Cross-Sell Different Products
# Even if spending is similar, purchase needs may differ:
# Married customers: groceries, household items, children's products.
# Unmarried customers: electronics, fashion, personal care.
# Walmart can recommend different products while maintaining similar promotional intensity.
# 4. Design Universal Loyalty Programs
# Since both groups have similar spending patterns, Walmart can create one loyalty program that rewards all customers based on spending rather than ma
# 5. Inventory Planning
# Marital status should not be used as a primary factor for forecasting revenue because average spending is nearly identical.
# Inventory decisions should instead be driven by age groups, product categories, and seasonal demand.
```

In [117…

```
## Age Analysis
df_walmart['Age'].unique()
```

Out[117…

```
<StringArray>
['0-17', '55+', '26-35', '46-50', '51-55', '36-45', '18-25']
Length: 7, dtype: str
```

In [118…

```
# Create Life Stage Groups

df_age = df_walmart.copy()
df_age['Age_Group'] = df_age['Age'].replace({
    '0-17':'0-17',
    '18-25':'18-25',
    '26-35':'26-35',
    '36-45':'36-50',
    '46-50':'36-50',
```

```
    '51-55':'51+',
    '55+':'51+'
})
```

In [119…
```
# Calculate Mean Purchase by Age Group
df_age.groupby('Age_Group')['Purchase'].mean()
```

Out[119…
```
Age_Group
0-17     8933.464640
18-25    9169.663606
26-35    9252.690633
36-50    9295.331743
51+      9463.661678
Name: Purchase, dtype: float64
```

In [120…
```
# Calculate 95% CI for Each Age Group
age_groups = df_age['Age_Group'].unique()
for group in age_groups:
    purchases = df_age[df_age['Age_Group']==group]['Purchase']
    lower, upper = confidence_interval(purchases,0.95)
    print(f"{group} : ({lower:.2f}, {upper:.2f})")
```

```
0-17 : (8851.95, 9014.98)
51+ : (9423.17, 9504.16)
26-35 : (9231.73, 9273.65)
36-50 : (9270.46, 9320.20)
18-25 : (9138.41, 9200.92)
```

In [121…
```
# Compare Age Groups
df_age.groupby('Age_Group')['Purchase'].agg(
    ['count','mean','std']
)
```

Out[121…

Age_Group	count	mean	std
0-17	15102	8933.464640	5111.114046
18-25	99660	9169.663606	5034.321997
26-35	219587	9252.690633	5010.527303
36-50	155714	9295.331743	5006.934324
51+	60005	9463.661678	5061.161476

In [122…
```
## Insights for Age Analysis :
# 1. Spending Increases with Age
# The average purchase amount generally increases as age increases.
# Younger customers spend less, while older customers spend more.
# 2. Highest Spending Group
# The 51+ age group has the highest average purchase amount:
```

```
# 9463.66
# This suggests that older customers have greater purchasing power and higher spending capacity.
# 3. Lowest Spending Group
# The 0-17 age group has the lowest average purchase amount:
# 8933.46
# This is expected because teenagers typically have limited income and purchasing authority.
# 4. Core Customer Segment
# The 26-35 and 36-50 groups account for the largest number of transactions.
# These age groups represent Walmart's primary customer base.
# 5. Consistent Spending Variability
# The standard deviation is around 5000 for all age groups.
# This indicates that purchase behavior variability is fairly similar across age segments.
```

In [123…

```
## Business Recommendations :
# For 0-17 Customers
# Promote affordable products.
# Offer student-oriented discounts and seasonal promotions.
# For 18-25 Customers
# Focus on fashion, electronics, and lifestyle products.
# Use digital marketing and app-based offers.
# For 26-35 and 36-50 Customers
# Target family-oriented products.
# Promote household essentials and bundled offers.
# For 51+ Customers
# Provide premium products and loyalty benefits.
# Focus on convenience, healthcare, and quality-focused offerings.
```

In [124…

```
# 7. Give recommendations and action items to Walmart.

# Recommendations and Action Items for Walmart
# 1. Focus Marketing Efforts on High-Spending Customer Segments

# The analysis shows that male customers have a significantly higher average purchase amount than female customers,
# and customers aged 51+ spend the most among all age groups.

# Action Items:

# Target high-value male customers with premium product promotions and bundle offers.
# Develop loyalty programs and exclusive deals for customers aged 51+.
# Increase marketing spend on segments that generate higher revenue.
# 2. Create Age-Based Marketing Campaigns

# Customer spending increases with age, with the 26-35, 36-50, and 51+ age groups contributing the highest average purchases.

# Action Items:

# Offer family-oriented promotions and household bundles for customers aged 26-50.
# Provide premium, health-related, and convenience-focused products for customers aged 51+.
# Launch student discounts and budget-friendly offers for younger customers (0-17 and 18-25).
# 3. Use Personalized Product Recommendations

# Different customer groups exhibit different spending behaviors and purchasing needs.
```

```
# Action Items:

# Implement recommendation systems based on customer demographics and purchase history.
# Use personalized coupons and targeted advertisements through the Walmart app and email campaigns.
# Promote complementary products through cross-selling and upselling strategies.
# 4. Prioritize Gender and Age Over Marital Status

# The confidence intervals for married and unmarried customers overlap significantly, indicating similar spending behavior.

# Action Items:

# Avoid creating separate pricing or promotional strategies solely based on marital status.
# Use gender, age, and purchasing patterns as primary segmentation variables.
# Allocate marketing resources toward segments with statistically meaningful spending differences.
# 5. Improve Customer Retention

# Retaining existing customers is often more cost-effective than acquiring new ones.

# Action Items:

# Expand loyalty and rewards programs.
# Offer personalized discounts to encourage repeat purchases.
# Identify high-value customers and provide exclusive benefits to increase retention.
# 6. Leverage Data Analytics for Business Decisions

# The analysis demonstrates that customer demographics influence spending patterns.

# Action Items:

# Continuously monitor purchasing behavior across demographic groups.
# Use predictive analytics to forecast demand and optimize inventory.
# Track campaign performance and refine targeting strategies based on customer response.
# Conclusion

# The analysis indicates that gender and age have a significant impact on customer spending, whereas marital status does not.
# Walmart should focus on age-based and gender-based marketing, personalized promotions, customer retention programs,
# and data-driven decision-making to improve customer engagement and maximize revenue growth.
```